

01.04.02 Прикладная математика и информатика



Network topologies and diffusion models

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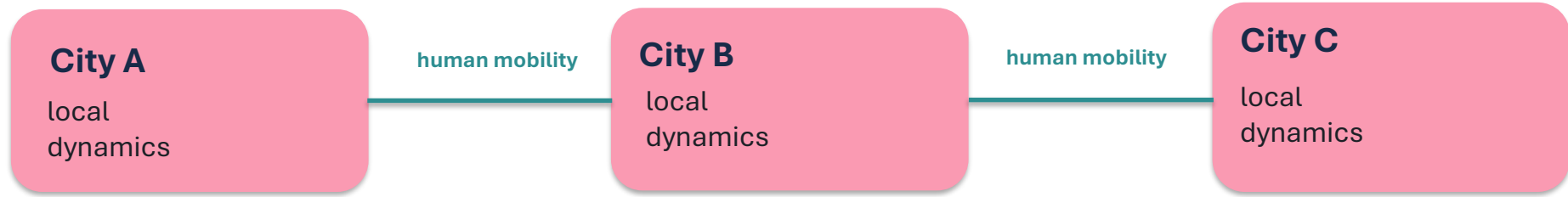
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Saint-Petersburg, 2026

Statement of the problem

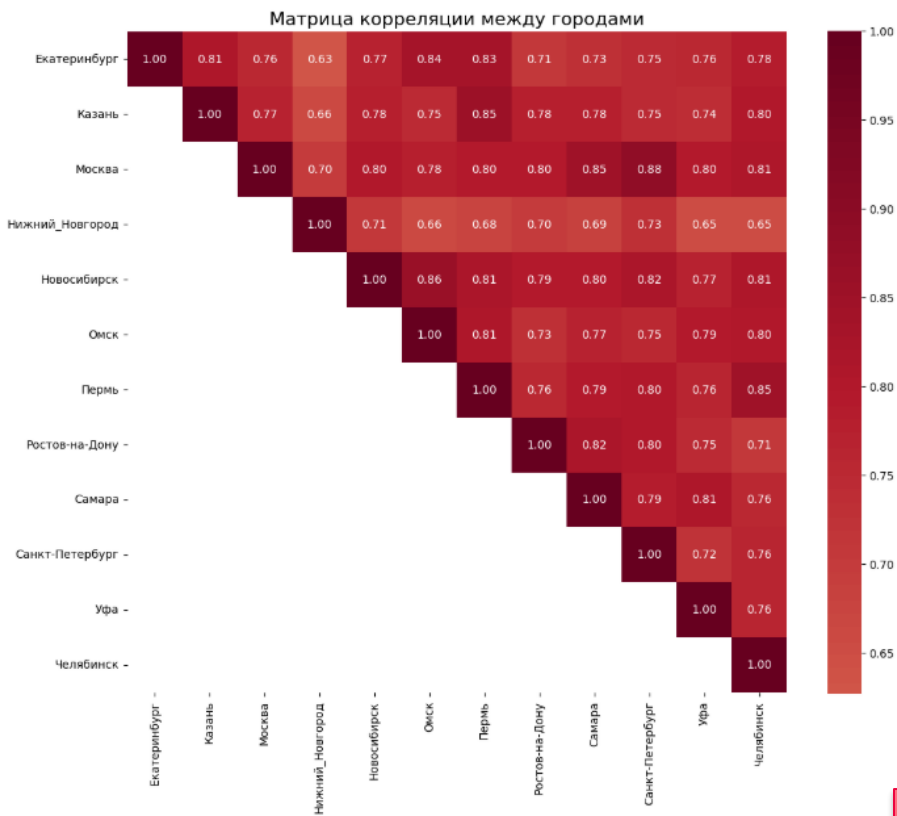
An epidemic doesn't develop independently in each city.



Problem

How important is it to account for inter-city links when predicting the timing of an epidemic peak?

Inter-city correlation in epidemic dynamics



- All 66 pairs of cities show a statistically significant correlation ($p < 0.01$).
- Strong connections ($r > 0.8$):
 - Moscow $\leftrightarrow$ Saint Petersburg: 0.883
 - Novosibirsk $\leftrightarrow$ Omsk: 0.862
 - Yekaterinburg $\leftrightarrow$ Perm: 0.835
- Weakest connection: Nizhny Novgorod $\leftrightarrow$ Chelyabinsk: 0.650

High synchronization confirms the need to take transport flows into account in the model.

Research hypothesis

If inter-city transport flows are taken into account,
the predicted timing of the epidemic peak
should become more accurate.

Incidence

Transport flows

Epidemic peak

Inter-city links

Disease spread

Previous research and research gap

Leonenko & Novoselova, 2018

«Influence of External Factors on Inter-City Influenza Spread in Russia: A Modeling Approach».

- inter-city influenza spread modeling
- inter-city migration included
- weekly incidence data
- assessment of model applicability for prediction

Focus of this study

How much can accounting for inter-city transport links improve the prediction of epidemic peak timing?

Core experiment



Without transport

vs.

With transport

Aim

To assess the effect of accounting for inter-city transport flows on the accuracy of epidemic peak timing prediction.

- 1 Investigate the relationship between epidemic dynamics across cities.
- 2 Formulate a hypothesis about the influence of inter-city transport links.
- 3 Build a disease-spread model incorporating transport flows.
- 4 Compare it with a model without transport links.

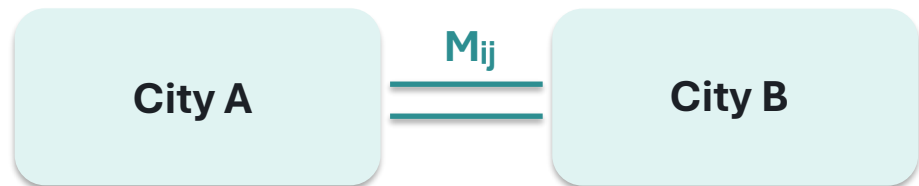
Inter-City disease spread model

Local dynamics within a city



$S \rightarrow I \rightarrow R$

Inter-city mobility



Target variable

Week of maximum incidence

How was the effect of transport tested?

Model A • With transport

$$\text{SIR} + M_{ij}$$

Cities are connected by population movement → peak prediction

Model A • With transport

$$\text{SIR} + M_{ij} \approx 0$$

Cities are treated as nearly independent → peak prediction

Compare the error in predicted peak timing

Data used in the model

1. Epidemiological data

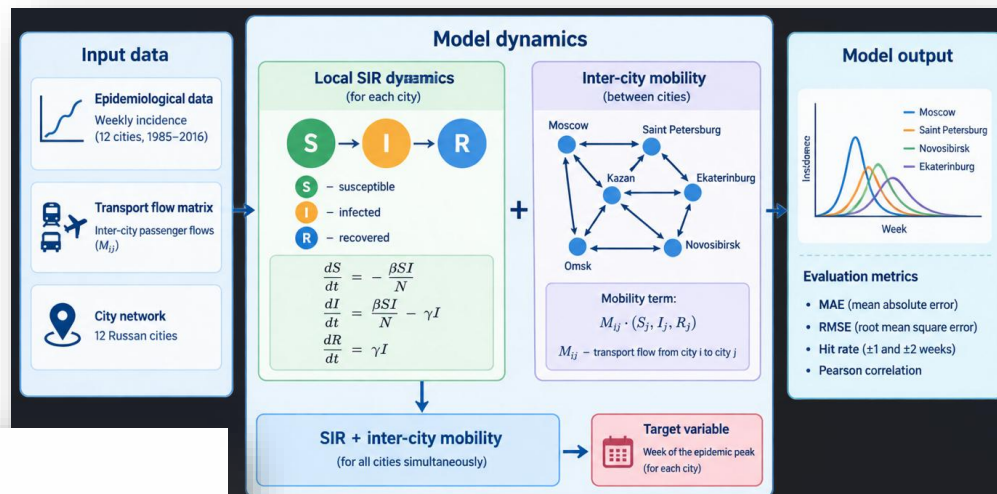
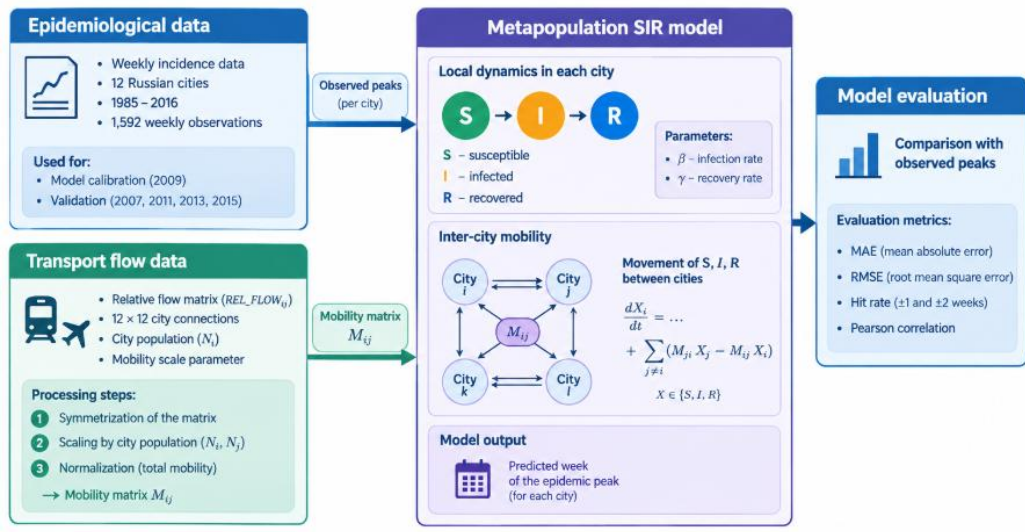
- Weekly disease incidence data
- **12 major Russian cities**
- Period: 1985-2016
- **1,592 weekly observations**
- Used to identify epidemic dynamics and determine the **observed epidemic peak week**

2. Inter-city transport data

- Inter-city passenger flow matrix M_{ij}
- Describes the relative intensity of movement between cities
- Used to model the movement of **susceptible (S), infected (I), and recovered (R)** populations between cities
- The transport matrix is scaled according to city population

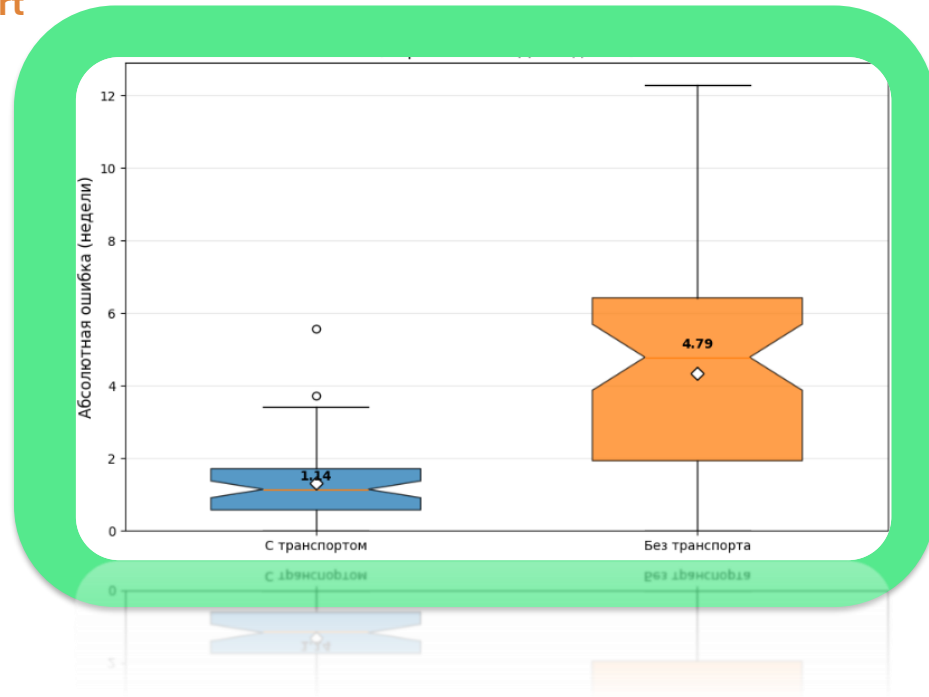
Inter-city transport data are represented by a relative mobility matrix, describing the intensity of population movement between each pair of cities. The matrix is scaled according to city population and a calibrated mobility parameter.

Block diagram of the model and data diagram



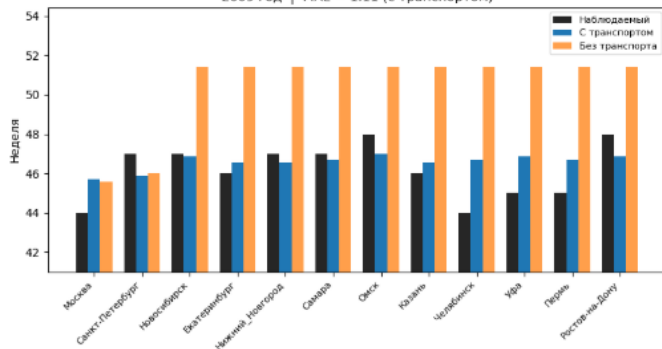
Interval estimation

Metric	With transport	Without transport
Median	1.14	4.79
Mean	1.32	4.35
Q1	0.57	1.93
Q3	1.71	6.43
CI	[0.79, 1.43]	[3.29, 5.43]

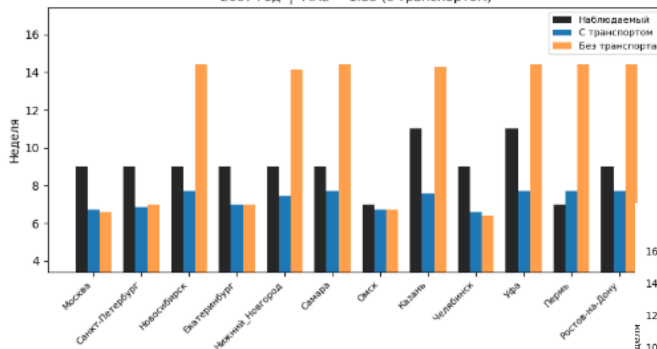


Prediction accuracy: with vs without transport

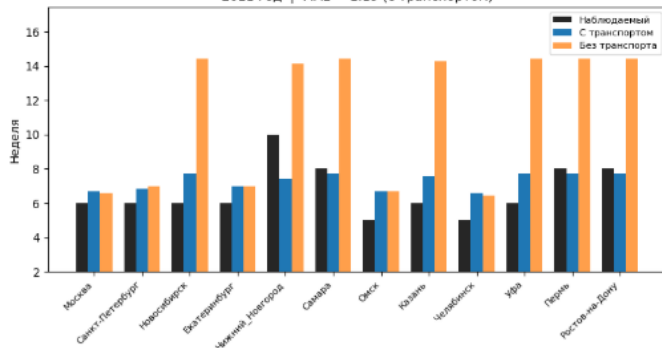
2009 год | MAE = 1.11 (с транспортом)



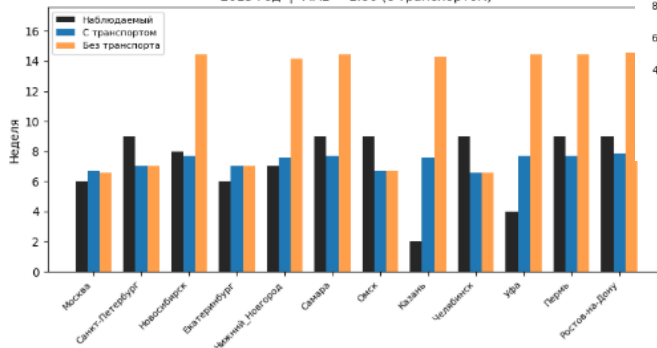
2007 год | MAE = 1.83 (с транспортом)



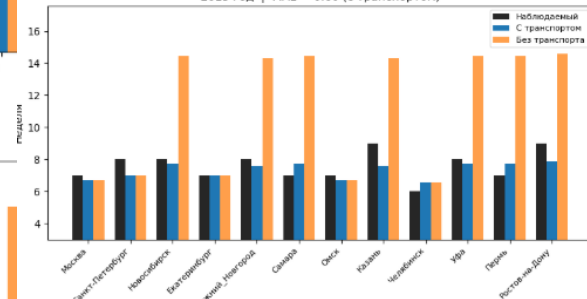
2011 год | MAE = 1.19 (с транспортом)



2013 год | MAE = 1.86 (с транспортом)



2015 год | MAE = 0.60 (с транспортом)



Example: epidemic peak prediction in 2015

Metric	With transport	Without transport
MAE	0.6	3.92
RMSE	0.72	4.94
Hit Rate ± 1 week	83.3%	41.7%
Hit Rate ± 2 weeks	100%	41.7%

Novosibirsk

Observed peak **Week 8**

With transport **7.7**
error = 0.3

Without transport **14.4**
error = 6.4

The transport component better synchronizes epidemic dynamics across cities.

Completion of the research objectives

- 1 The relationship between the dynamics of epidemics in different cities has been investigated.
- 2 A hypothesis has been formulated regarding the impact of intercity transport links.
- 3 A disease spread model has been developed that takes into account transport flows.
- 4 A comparison was made with a model without transport links.

Thank you for attention!

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